Does the Depth-Camera Model Change the Grasp? A Validated RealSense D415 Simulation for Aluminium Profiles

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***Abstract*—Synthetic RGB-D data are increasingly used to develop robotic perception, yet rendered depth lacks the artefacts that active-stereo cameras produce on specular metal. We simulate the Intel RealSense D415 in Isaac Sim for scenes of aluminium profiles and brackets and compare five depth models of increasing realism, from ideal depth to a physics-based model that renders the dot projector and computes depth by census transform and semi-global matching. Against the real D415 depth of metal parts, the physics-based model reproduces the disparity-error distribution on anodized aluminium. In a paired grasp experiment on 1,075 profiles, ideal and Gaussian depth predict 100% success, whereas the physics-based model predicts 44–85% on polished profiles (33–85% across its threshold and sensor variants; McNemar test with Holm correction) and is the only model that exposes a failure mechanism: disparity propagation across textureless faces parallel to the stereo baseline (odds ratio 0.04–0.11). A depth-only segmentation network trained on ideal depth drops from 85% to 11% intersection-over-union on simulated D415 depth. The predicted grasp success, however, ranges from 39% to 96% with the roughness of the polished finish, a parameter we could not measure. Code and results: https://github.com/davidepillon001/isaac-sim-realsense-metal.**

***Index Terms*—RGB-D sensor simulation, active stereo, sim-to-real, robotic grasping, reflective objects, synthetic data.**

I. Introduction

Robots that pick and assemble aluminium profiles perceive them with RGB-D cameras, and their perception is increasingly trained and validated on synthetic data. A renderer, however, returns *ideal* depth, whereas an active-stereo camera loses depth or reports wrong depth precisely on shiny aluminium: a specular highlight moves with the viewpoint, the two infrared views no longer show the same texture, and the matcher either rejects the pixel or accepts a false correspondence. A simulated camera that is cleaner than the real one can therefore validate a pipeline that fails on the robot.

We take the Intel RealSense D415 [1] as reference sensor: an active-stereo camera with baseline $B = 55\,\text{mm}$, a near-infrared dot projector and two 1280×720 infrared imagers. We chose it because two public datasets provide *raw* D415 depth of metal parts with reference geometry [2], [3], so that the simulated sensor can be validated against the real one rather than against its datasheet.

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We ask whether the depth-camera model changes the outcome of a downstream perception or grasp step and, if so, through which mechanism. The design is paired: the same 200 scenes are processed with depth models of increasing realism and the same estimator is evaluated on the same 1,075 profiles. Our contributions are:

1) an Isaac Sim pipeline that renders RGB, a D415 infrared pair lit by a simulated dot projector, and depth, segmentation, normal and pose labels (Sec. III);
2) an empirical artefact model calibrated per incidence angle on real D415 data and a physics-based stereo model, both validated against real depth of metal parts with standard stereo metrics (Sec. V-B);
3) a paired grasp and segmentation study with scene-clustered statistics that quantifies the effect of the camera model and identifies the failure mechanism and its sensitivity to the material (Sec. V-C–Sec. V-F).

II. Related Work

Depth-sensor simulation. DepthSynth [4] renders the projected pattern of structured-light sensors and block-matches it across the projector–camera baseline, and shows that the resulting depth improves recognition trained on synthetic data. Zhang et al. [5] simulate the D415 with ray-traced infrared images of the measured dot pattern, census transform, semi-global matching [6] and consistency checks, and quantify the sim-to-real gap through the correlation of nine perception models on aligned real and simulated scenes; DREDS [7] uses a similar simulator to train depth restoration for specular and transparent objects. These works validate at the *task level*, by correlating the performance of downstream models on aligned real and simulated scenes [5], [8]. We validate at the *error level*: the distribution of the depth error of the simulated sensor is compared with that of the real one on independent datasets, and the task level is used only to expose its consequences. Grasp learning often relies on simpler noise: Dex-Net 2.0 [9] augments depth with Gamma and Gaussian-process noise, whose removal lowered physical grasp success from $83\pm8\%$ to $78\pm9\%$, whereas DiffuDepGrasp [10] reports 0% real-camera success for a policy trained on clean or randomly perturbed depth and 95.7% with a learned noise model.

Real depth of reflective parts. ROBI [2] provides multi-view D415 depth of reflective industrial parts at about 0.35 m

with Ensenso reference depth of the sprayed parts; XYZ-IBD [3] provides raw D415 depth of 15 metal parts at a sensor-to-object distance of 0.6–1.0 m with CAD models and 6-D poses. ActiveZero [11] reports a D415 depth error of 15.8 mm on specular and translucent objects, almost three times the 5.8 mm scene average.

III. Simulation of the D415

A. Scenes and rendering

A T-slot profile of section 40×40 mm, with a central core and four slots, is extruded to a length $L \sim \mathcal{U}(0.12, 0.40)$ m; an L-bracket measures 40×40×4 mm. Each scene contains 4–10 profiles and 2–6 brackets placed on a table by rejection sampling without interpenetration. Every part is anodized (metallic 1.0, roughness 0.45) or polished aluminium (metallic 1.0, roughness 0.10) in a physically based material model. The camera observes the table from a distance $z_c \sim \mathcal{U}(0.6, 1.2)$ m at an elevation of 50–80°. We render 200 scenes with a fixed seed; a profile enters the experiments if it has at least 800 visible pixels and does not touch the image border, which yields 563 anodized and 512 polished profiles.

The virtual D415 follows the datasheet: 1280 × 720 px, horizontal field of view 65°, hence $f_x = 640/\tan 32.5° = 1004.6$ px, baseline $B = 55$ mm and RGB sensor 15 mm from the left infrared imager. Since the renderer does not support textured light sources, the dot projector is modelled, as a spotlight cookie in DepthSynth [4], by a point light behind a perforated mask (600×600 cells, 8% open, 40° cone) between the two imagers. Images are path-traced with 32 samples per pixel, and depth, semantic and instance masks, normals and poses come from the Replicator annotators (Fig. 1). Rendering takes about 12 s per scene on a laptop NVIDIA RTX GPU (200 scenes with seed 1 in about 40 min); all other steps run on a laptop CPU.

B. Depth models

For a rectified pair a point at depth z has disparity $d = f_x B/z$, with $f_x B = 55.25$ px m. To first order a disparity error δd causes

$$\delta z = -\frac{z^2}{f_x B}\,\delta d, \qquad \sigma_z = \frac{z^2}{f_x B}\,\sigma_d, \tag{1}$$

so a constant matching error in pixels grows quadratically in depth, as measured for the RealSense D435 [12]. All models therefore act on disparity.

Ideal and Gaussian. The ideal model returns the rendered depth. The Gaussian model adds $\delta d \sim \mathcal{N}(0, \sigma_d^2)$ with $\sigma_d = 0.16$ px, twice the laboratory value of Intel [1], i.e. $\sigma_z = 1.0$ mm at 0.6 m and 4.2 mm at 1.2 m, and never removes a pixel: it is the usual datasheet-level model.

Heuristic artefact model. It implements the artefact classes of DepthSynth [4] with hand-set parameters: noise $\sigma_d/\max(\cos\theta, 0.2)$, a hole probability rising from 0 to 1 for incidence angles θ of 70–85°, holes at depth discontinuities and occlusions, correlated blobs of holes (5% anodized, 30% polished pixels), saturation near the specular direction and $1/16$ px quantisation, but no false matches.

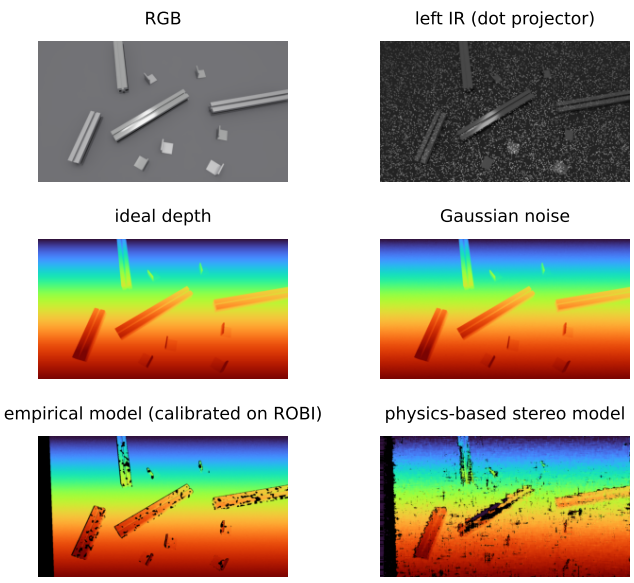


Fig. 1: A synthetic scene: RGB, left infrared image and depth under four models. Polished faces barely return the projected dots; the physics-based model produces holes and false matches that follow the reflections, the empirical model blobs drawn from correlated random fields, the Gaussian model none.

Empirical artefact model. The hand-set rates are replaced by statistics of real D415 depth of metal (ROBI Zigzag Scene 4 for anodized, Chrome Screw Scene 4 for polished aluminium). For each incidence-angle bin b of 15° we estimate the probability of a hole $p_h(b)$ and of a wrong pixel $p_w(b)$, i.e. a valid pixel with $|\delta d| > 1$ px, the empirical distribution $\hat{F}_\Delta$ of the errors of wrong pixels, and the noise of the remaining pixels by $\hat{\sigma}_d = 1.4826\,\mathrm{MAD}$ (0.52 px on Zigzag, 0.51 px on Chrome). Holes and wrong pixels are sampled by thresholding Gaussian random fields so that they form blobs, and wrong pixels draw a spatially coherent error from $\hat{F}_\Delta$. Since ROBI is recorded at 0.35 m and our scenes at 0.6–1.2 m, an error $\Delta \sim \hat{F}_\Delta$ must be transferred across distance; two rules are physically plausible,

$$\hat{z} = z + \Delta_{\mathrm{mm}} \quad \text{(metric)}, \qquad \hat{z} = \frac{f_x B}{f_x B/z + \Delta_{\mathrm{px}}} \quad \text{(disparity)}. \tag{2}$$

The metric rule holds if a false match follows the reflection geometry, the disparity rule if it is a matching error of fixed size in pixels; we keep both (Sec. V-G).

Physics-based stereo model. Depth is computed from the rendered infrared pair with a matching chain modelled on the D415, as in [5]. A 9×9 centre-symmetric census transform [13] maps each pixel to a bit string $\mathbf{c}(\mathbf{p})$, the matching cost is the Hamming distance $C(\mathbf{p}, d) = \mathrm{H}\big(\mathbf{c}_L(\mathbf{p}), \mathbf{c}_R(\mathbf{p} - d\,\mathbf{e}_x)\big)$, and four-path semi-global matching [6] with penalties $P_1 = 15$, $P_2 = 120$ aggregates it into $S(\mathbf{p}, d)$, with

$d^\star(\mathbf{p}) = \arg\min_d S(\mathbf{p}, d)$. A pixel is rejected if the second-best cost is within 10% of the best, if left–right consistency fails by more than 1 px, or if it deviates by more than 1 px from its 3×3 median. A Lucas–Kanade step over a 15×15 window refines the integer disparity. Holes and false matches thus *emerge* from the rendered images and depend on material, viewpoint, projector texture and orientation, with no rate set by hand. Two variants test robustness: OpenCV SGBM with strict uniqueness (30%) and texture thresholds, and census matching on infrared images formed from linear radiance with auto-exposure and thermal noise (HDR sensor).

Modelled and omitted effects. The physics-based model includes the effects that decide *where* matching fails on metal: the projector texture, specular reflection and inter-reflections (path tracing), census matching with its consistency checks, sub-pixel noise and pixel locking (calibrated in Sec. V-A), and occlusion holes, which follow from the two-view geometry. Three effects were implemented and rejected because they increased the distance to the real data (Sec. VI-A): the measured infrared noise of the D415, laser speckle and the HDR sensor. Intel's post-processing filters are omitted because the reference datasets use raw depth; rolling shutter and motion blur because scene and camera are static; temporal noise, thermal drift, lens distortion and calibration error because single rectified frames are used and no data were available to calibrate them; ambient infrared because the cell is indoors.

IV. Experimental Protocol

Real reference data. From ROBI [2] we use Scenes 4 and 5 of Zigzag (zinc-plated) and Chrome Screw, up to 125 views each. The reference is Ensenso N35 depth of the sprayed parts (accuracy about 0.2 mm) reprojected into the D415 frame. Metal pixels lie within 25 mm of an annotated object centre, eroded by 3 px. From XYZ-IBD [3] we use 150 frames of the 15 validation scenes, with reference depth obtained by ray casting the CAD models at the annotated poses; the nominal sensor-to-object distance is 0.6–1.0 m, and the metal pixels lie at optical-axis depths $z = 0.5$–0.85 m.

Depth metrics. On metal pixels we evaluate the disparity error $e = f_x B/\hat{z} - f_x B/z$, which is comparable across distances by Eq. (1), with the Middlebury v3 metrics [14]: invalid rate, $\mathrm{bad}_t = \frac{1}{N}\sum_i \mathbb{1}[|e_i| > t]$ for $t \in \{1, 2\}$ px and quantiles A_q of $|e|$; errors are computed on valid pixels and invalid pixels are counted separately [11]. Distributions are compared by

$$W_1 = \int_0^\infty \big|F_\mathrm{s}(x) - F_\mathrm{r}(x)\big|\,dx, \quad D_\mathrm{KS} = \sup_x \big|F_\mathrm{s}(x) - F_\mathrm{r}(x)\big|, \tag{3}$$

where F_s, F_r are the distributions of $|e|$ in simulation and on the matching real data (anodized: ROBI Zigzag and XYZ-IBD; polished: ROBI Chrome). With millions of pixels every difference is significant, so we report effect sizes. Following [8], the Sim-vs-Real Correlation Coefficient (SRCC) is the Pearson correlation r between simulated and real hole and wrong-pixel rates over the six angle bins (12 points), with Fisher interval $\tanh\big(\operatorname{artanh} r \pm 1.96\,(n-3)^{-1/2}\big)$.

Grasp task. For each profile the depth inside its ground-truth instance mask is back-projected to heights h_i above the table, so that only depth errors matter. The top-face height is estimated by the *percentile* estimator $\hat{h} = Q_{0.9}(\{h_i\})$ or by the *mode* estimator $\hat{h} = \arg\max_h (\mathcal{G}_{2\,\mathrm{mm}} * \mathrm{hist}_{1\,\mathrm{mm}})(h)$. The axis is the first principal component of the top-face points and the lateral centre the midpoint of their 2–98% extent. A grasp succeeds if the lateral and height errors are below $\tau = 10$ mm, the angle error below 10° and the error along the axis below 30 mm, the tolerance of a parallel gripper with 70 mm opening on a 40 mm profile. The percentile estimator ignores depth below the surface but is biased upwards by noise; the mode estimator is unbiased under symmetric noise but follows the largest coherent cluster of wrong depth. For independent noise the percentile bias admits a closed form:

Proposition. If the disparity of the top-face points carries i.i.d. noise $\mathcal{N}(0, \sigma_d^2)$, then as the number of points grows $\hat{h} - h \to \Phi^{-1}(0.9)\,\sigma_z = 1.28\,z^2\sigma_d/f_x B$ by Eq. (1), and points below the face can only lower $Q_{0.9}$. Hence noise alone makes the percentile estimator exceed a height tolerance τ only beyond

$$z^\star = \sqrt{\tau\, f_x B \,/\, \big(\Phi^{-1}(0.9)\,\sigma_d\big)}. \tag{4}$$

Statistical analysis. Profiles of one scene share viewpoint and lighting and are not independent, so inference resamples or models scenes. For each model and finish we report $\hat{p} = \sum_j k_j / \sum_j n_j$, with k_j successes among the n_j profiles of scene j, a 95% interval from 10^4 cluster-bootstrap resamples of scenes [15], and the design effect $\mathrm{Var}_\mathrm{boot}(\hat{p})/[\hat{p}(1-\hat{p})/n]$. Two models are compared on the same profiles by the exact McNemar test: if n_{10} profiles succeed only with the first and n_{01} only with the second, $n_{10} \sim \mathrm{Bin}(n_{10} + n_{01}, \frac{1}{2})$ under H_0 [16]; the 18 p-values per estimator (9 pre-specified model pairs × 2 finishes, listed in the repository) are adjusted by Holm's procedure [17]. Determinants of failure on polished profiles are estimated by a logistic GEE with exchangeable within-scene correlation [18],

$$\mathrm{logit}\,P(s_{ij} = 1) = \beta_0 + \beta_\parallel \cos^2\phi_{ij} + \beta_z\,\Delta z_{ij} + \beta_L\,\Delta L_{ij}, \tag{5}$$

where ϕ_{ij} is the angle between profile i of scene j and the baseline, and Δz, ΔL are the centred distance and length in units of 10 cm, so that e^β is an odds ratio. The regressor $\cos^2\phi$ is the fraction of the gradient energy of the profile edges that is lost along the epipolar direction (Eq. (6)). The GEE is exploratory: its p-values are not adjusted for multiplicity.

Additional experiments. (i) The first 30 scenes are re-rendered with polished roughness 0.05, 0.20 and 0.30 and identical geometry. (ii) A U-Net (4 levels, 16 base channels) segments background, profiles and brackets from depth alone (320 × 180 px; inputs: height relative to the image median and validity mask); it is trained on scenes 0–159 with one depth model and tested on scenes 160–199 with another, with 3 seeds, and evaluated by the profile intersection-over-union $|P \cap G|/|P \cup G|$ and the recall $|P \cap G|/|G|$ per finish.

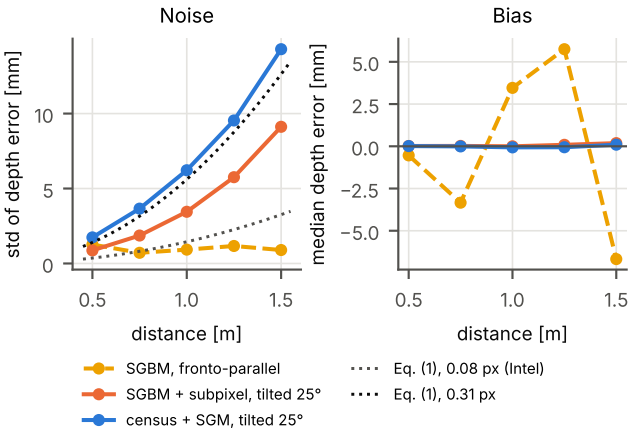


Fig. 2: Stereo matching on a textured plane: noise (left) and median error (right) of the depth vs. distance. Dotted lines: Eq. (1) with $\sigma_d = 0.08$ px (Intel) and 0.31 px. SGBM on a fronto-parallel plane shows pixel locking.

V. Results

A. Calibration on a plane

Sub-pixel interpolation of a matching cost is biased towards integer disparities (pixel locking). On a fronto-parallel plane all pixels share the same fractional disparity, so the bias does not average out: the median error of SGBM oscillates by up to ± 7 mm with distance while its spread appears smaller than the datasheet value (Fig. 2). On a plane tilted by 25° the fractional parts are spread over $[0, 1)$ and, after one Lucas–Kanade step, the bias falls below 0.5 mm. The census chain then has a constant disparity noise $\hat{\sigma}_d \approx 0.31$ px and, by Eq. (1), a quadratically growing σ_z. This is four times the 0.08 px quoted for laboratory conditions [1] but consistent with the real camera in a scene: on the matte table of ROBI the median $|e|$ is 0.19 px, against $\Phi^{-1}(0.75)\,\hat{\sigma}_d = 0.21$ px expected for $e \sim \mathcal{N}(0, \hat{\sigma}_d^2)$.

B. Depth error on metal: simulation vs. real D415

Tab. I and Fig. 3 lead to four observations. (i) *Gaussian noise cannot produce the failure mode*: with $\sigma_d = 0.16$ px, $P(|e| > 1\,\text{px}) = 2[1 - \Phi(6.25)] < 10^{-9}$, whereas the real D415 has bad_1 of 8–26% on metal, and its distribution is the farthest from every real set ($W_1 = 0.34$–0.64 px). (ii) *On anodized aluminium the physics-based model matches XYZ-IBD*, the real data closest to our working distance: bad_1 8.2% vs. 8.3%, equal A_{90}, $W_1 = 0.10$ px and $D_{\text{KS}} = 0.03$. (iii) *On polished aluminium it overestimates the tail*: bad_1 36.5% vs. 26.0% and A_{99} 38 px vs. 4 px on real chrome, so it probably overestimates grasp failures on polished parts. (iv) *The empirical model is closest to ROBI* ($W_1 = 0.07$–0.12 px with disparity transfer), partly by construction; on the held-out Scene 5 its angle curves deviate by 2.5/1.1 points (Zigzag, holes/wrong) and 2.9/2.3 points (Chrome), against 8.7/15.0 and 39.3/24.6 for the heuristic model.

TABLE I: Disparity error on metal pixels (Middlebury metrics on valid pixels) and Wasserstein distance W_1 of $|e|$ to the matching real data. Bold: closest model.

Source	inval. [%]	bad1 [%]	bad2 [%]	A_{50} [px]	A_{90} [px]	A_{99} [px]	W_1 [px] ROBI	W_1 [px] XYZ
Real D415								
ROBI Zigzag (0.35 m)	14.7	16.5	5.4	0.43	1.33	7.34	–	–
ROBI Chrome (0.35 m)	10.7	26.0	4.9	0.58	1.54	3.96	–	–
XYZ-IBD (z 0.5–0.85 m)	1.9	8.3	2.1	0.31	0.91	2.81	–	–
Simulated, anodized aluminium								
Gaussian noise	0.0	0.0	0.0	0.11	0.26	0.41	0.60	0.34
Heuristic	13.8	1.2	0.2	0.14	0.40	1.08	0.53	0.27
Empirical, metric	19.0	6.1	0.5	0.34	0.86	1.53	0.30	**0.10**
Empirical, disparity	19.0	19.7	3.5	0.43	1.28	4.25	**0.12**	0.19
Physics-based stereo	9.0	8.2	1.9	0.28	0.91	2.60	0.38	**0.10**
Simulated, polished aluminium								
Gaussian noise	0.0	0.0	0.0	0.11	0.26	0.41	0.64	–
Heuristic	43.0	1.2	0.2	0.15	0.42	1.08	0.56	–
Empirical, metric	15.3	4.8	0.3	0.32	0.81	1.42	0.36	–
Empirical, disparity	15.3	27.2	4.5	0.48	1.51	3.75	**0.07**	–
Physics-based stereo	30.4	36.5	16.8	0.67	2.80	37.98	1.04	–

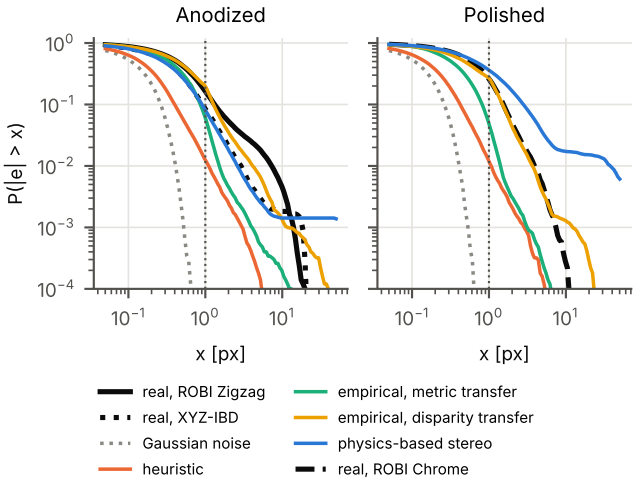


Fig. 3: Exceedance $P(|e| > x)$ of the disparity error on valid metal pixels; the value at $x = 1$ px is bad_1.

In all three real sets the wrong-depth rate grows with the incidence angle (Fig. 4; XYZ-IBD 5→21%, Zigzag 4→27%, Chrome 13→37%), as the specular lobe turns away from the camera and the projected texture weakens. The physics-based model reproduces this trend on anodized aluminium (SRCC with Zigzag $r = 0.70$, 95% CI [0.21, 0.91]) but not on polished aluminium ($r = -0.04$). With XYZ-IBD $r = 0.13$, since its nearly constant hole rate (1–3%) leaves little variance, although the curves differ by only 5.7 points on average. Most wrong pixels of the real camera lie *behind* the surface (Zigzag 73%, Chrome 80%, XYZ-IBD 54%), as expected for a virtual image formed by a mirror; the physics-based model reproduces this sign (73% anodized, 94% polished).

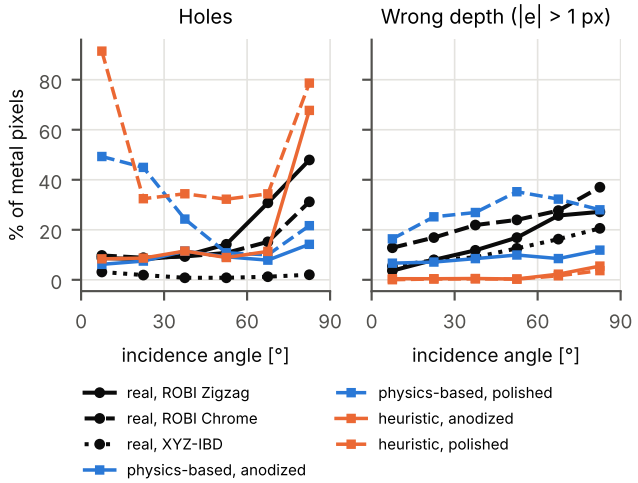


Fig. 4: Holes and wrong depth on metal vs. incidence angle: real D415 (black) against the physics-based and heuristic models.

TABLE II: Grasp success [%] with 95% scene-bootstrap intervals (563 anodized, 512 polished profiles in 200 scenes). The last two rows are the robustness variants of the physics-based model (Sec. III-B).

	Anodized		Polished	
Depth model	percentile	mode	percentile	mode
Ideal	100	100	100	100
Gaussian noise	100	100	100	100
Heuristic	100	100	92.4 [89.9, 94.7]	92.2 [89.6, 94.6]
Empirical, metric	80.5 [75.1, 85.6]	100	91.8 [88.2, 95.1]	100
Empirical, disparity	68.0 [62.2, 73.7]	99.5	65.4 [59.6, 71.1]	96.5 [94.8, 98.0]
Physics-based stereo	99.8	99.8	**85.2** [82.0, 88.2]	**44.5** [40.0, 49.1]
SGBM, strict thresholds	100	99.8	77.0 [72.8, 80.9]	36.5 [32.0, 41.1]
HDR sensor	99.6	99.8	82.4 [78.9, 85.8]	32.8 [28.7, 36.8]

C. Effect of the camera model on the grasp

With ideal or Gaussian depth every profile is grasped (Tab. II). This follows from Eq. (4): with $\sigma_d = 0.16\,\text{px}$ and $\tau = 10\,\text{mm}$, $z^\star = 1.64\,\text{m}$ lies beyond the workspace (the bias is at most 5.4 mm at 1.2 m), and the mode estimator is unbiased. Under the physics-based model polished profiles succeed in 85.2% (percentile) and 44.5% (mode) of cases; on the same profiles the Gaussian model is optimistic by 14.8 [11.8, 18.0] and 55.5 [50.9, 60.0] points ($p_{\text{Holm}} < 0.01$; for the mode estimator $n_{10} = 284$ profiles succeed only with Gaussian depth and $n_{01} = 0$ only with the physics-based model). The conclusion does not depend on the tolerance: with $\tau = 5$ and 15 mm the gap is 36.9 and 6.1 points (percentile) and 78.3 and 40.6 points (mode), always with $n_{01} = 0$ and $p < 10^{-9}$. Anodized profiles succeed in 99.8% of cases, consistent with Sec. V-B.

The empirical model fails for a different reason. Its valid pixels carry independent noise $\hat{\sigma}_d \approx 0.52\,\text{px}$, so by Eq. (4) $z^\star \approx 0.91\,\text{m}$: the percentile estimator fails in the far half of the workspace even on anodized parts (80.5%), whereas the mode estimator does not (100%). On polished parts the two transfer rules bracket the physics-based model: +6.6 [2.0, 11.2] points with metric transfer ($p_{\text{Holm}} = 0.007$) and −19.7 [−26.7, −12.9] with disparity transfer (percentile estimator).

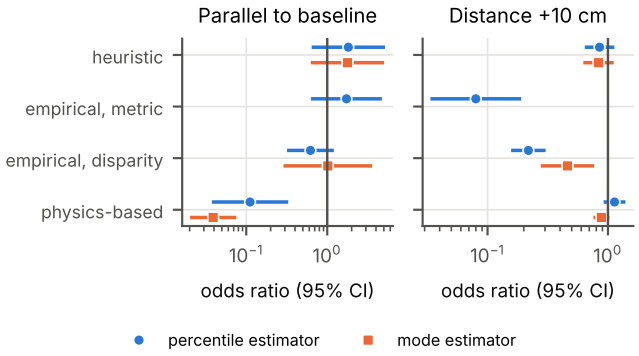


Fig. 5: Odds ratios of grasp success on polished profiles from the GEE model Eq. (5).

The two estimators disagree because they fail under different conditions. The percentile estimator fails only if fewer than 10% of the points lie within tolerance of the top face, since depth below the surface does not move $Q_{0.9}$; the mode estimator fails as soon as a coherent patch of wrong depth outnumbers the correct top-face points. The large coherent false matches on polished faces (Fig. 3) meet the second condition far more often than the first, and 87% (percentile) and 99% (mode) of the failures are height errors. The design effect is 1.0–1.2 for the physics-based model and 2.0–2.6 for the empirical model with the percentile estimator, whose failures concentrate in the far scenes (Eq. (4)).

D. Failure mechanism: the aperture problem along the baseline

Under the physics-based model a profile parallel to the baseline has an odds ratio of success of 0.11 [0.04, 0.31] (percentile) and 0.04 [0.02, 0.07] (mode) relative to a perpendicular one (Fig. 5); for $\phi < 30°$ the success falls to 72% and 15%. The heuristic and empirical models show no such effect (odds ratios 0.6–1.8, intervals including 1), since their artefacts do not depend on the baseline. Fig. 6 shows the cause. The polished top face reflects the projected dots away from the camera and is nearly textureless in both infrared images. In a rectified pair disparity is constrained only by intensity variation *along* the epipolar lines, i.e. along the baseline (the aperture problem of stereo). Quantitatively, for a photometric matching cost with image noise σ_n the Fisher information on the disparity of a window $\mathcal{W}$ is the energy of the gradient along the epipolar direction [19]; if the texture consists of edges along the profile axis, their gradient is normal to the axis and has component $|\nabla I| \sin\phi$ along the baseline, so that

$$\mathcal{I}_d = \frac{1}{\sigma_n^2} \sum_{\mathcal{W}} \left(\frac{\partial I}{\partial x}\right)^2 = \frac{\sin^2\phi}{\sigma_n^2} \sum_{\mathcal{W}} |\nabla I|^2, \quad \operatorname{Var}(\hat{d}) \geq \mathcal{I}_d^{-1}. \tag{6}$$

The census transform replaces the Gaussian noise model by bit flips and changes the constant, but preserves the dependence

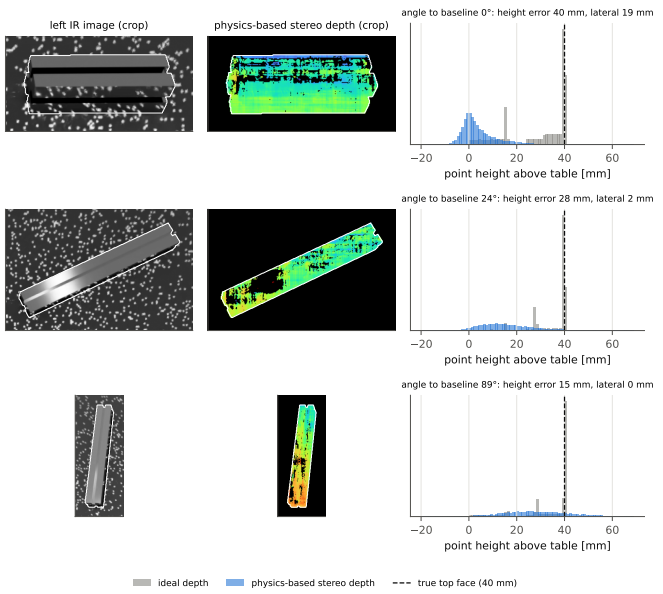


Fig. 6: Polished profiles grasped with ideal depth and missed under the physics-based model (mode estimator): most parallel, median and most perpendicular to the baseline. Left: infrared image; centre: depth; right: point heights. The parallel profile collapses to the table plane.

on the gradient energy along the epipolar direction. As $\phi \to 0$ the information vanishes and the bound diverges, $C(\mathbf{p}, d)$ becomes flat in d on the textureless top face, and semi-global aggregation propagates the disparity of the table into the face, which is measured at table height (40 mm error in Fig. 6). A perpendicular profile is crossed by its own edges, which anchor the correct disparity. Conversely, only the empirical model predicts a strong decrease of success with distance (odds ratio 0.08 [0.03, 0.18] per 10 cm, metric transfer), consistent with Eq. (4); the physics-based model shows none with the percentile estimator (1.13 [0.95, 1.35]) and a weak one with the mode estimator (0.88 [0.78, 1.00], unadjusted $p = 0.04$). The orientation effect is a cheap, falsifiable prediction for a real robot.

E. Sensitivity to the material

The polished roughness (0.10) was set by hand. Re-rendering the same 30 scenes with roughness 0.05–0.30 changes the prediction of the physics-based model more than any choice of sensor model (Fig. 7a): from 72.6% to 98.8% with the percentile and from 39.3% [27.2, 52.3] to 96.4% [92.5, 100] with the mode estimator, while ideal depth remains at 100%. The mechanism is the one above: a rougher surface has a wider specular lobe, returns part of the projected dots to the camera and restores texture on the top face. At roughness 0.05 parallel profiles succeed in 9% of cases with the mode estimator; at 0.30 the orientation effect almost vanishes. Distance matters far less than material and model (Fig. 7b,c), except for the empirical model with the percentile estimator, as predicted by Eq. (4).

F. Learned perception: train on ideal, test on realistic depth

TABLE III: Depth-only segmentation on 40 held-out scenes: profile IoU and recall per finish [%], mean (standard deviation) over 3 seeds.

Train on	Test on	IoU	Recall anod.	Recall pol.
Ideal	Ideal	85.0 (7.9)	99.5	99.4
Ideal	Phys. stereo	**10.9** (1.4)	75.0 (3.3)	48.3 (9.1)
Gaussian	Phys. stereo	13.4 (1.9)	85.5 (7.1)	55.4 (8.6)
Phys. stereo	Phys. stereo	71.2 (0.5)	98.2	95.3
Phys. stereo	Ideal	82.3 (0.8)	94.5	92.5

A network trained on ideal depth collapses on physics-based depth (Tab. III: IoU 85.0→10.9%). Recall remains at 48–75%, so the collapse consists mostly of false positives: holes and streaks on the table, absent from the training distribution, are labelled as profiles. Gaussian augmentation does not help (13.4%): it creates neither holes nor coherent false surfaces. Training on physics-based depth recovers 60.3 [56.5, 63.7] IoU points (paired bootstrap over test scenes and seeds; polished recall +47 [36, 62] points), loses only 2.7 points on ideal depth and is more stable across seeds (standard deviation 0.5 vs. 7.9 points). This agrees with [10]. The test is synthetic: it shows that the camera model decides whether a network trained in simulation can work at all, not how well it works on the real D415.

G. Transferring real errors across distance

The transfer rules Eq. (2) lead to different predictions (Tab. II), so we tested them on XYZ-IBD (pixel depths z = 0.5–0.85 m). If a quantile q of $|\hat{z} - z|$ scales as z^{α}, then $\alpha = \ln(q_2/q_1)/\ln(z_2/z_1)$, with $\alpha = 0$ under the metric and $\alpha = 2$ under the disparity rule. Between the bins centred at 575 and 800 mm the 90th percentile grows from 7.8 to 10.0 mm ($\alpha \approx 0.75$) and the 97th from 14.5 to 24.0 mm ($\alpha \approx 1.5$). Neither rule holds: moderate errors behave like reflection geometry, the tail like matching errors. Since the reference depth derives from annotated poses with millimetre-level error, which biases α towards 0, we report the empirical model as a range.

VI. Discussion

A. What did not work

1) **Projector as a textured area light.** Isaac Sim does not project textures of area lights, a limitation that Zhang et al. removed by extending their renderer [5]; we used a point light behind a perforated mask, enlarged to cover the field of view, with a 40° cone against light leakage.
2) **OpenCV SGBM.** It left the first $d_{\min} + d_{\text{num}}$ columns empty (fill rate 91–94%, fixed by padding) and showed pixel locking (Fig. 2); iterating the sub-pixel refinement increased the disparity noise at 1 m from 0.31 to 0.38 px. We therefore implemented census matching and semi-global matching ourselves.
3) **Additional sensor physics.** On 3 scenes the measured infrared noise of the D415 and laser speckle increased

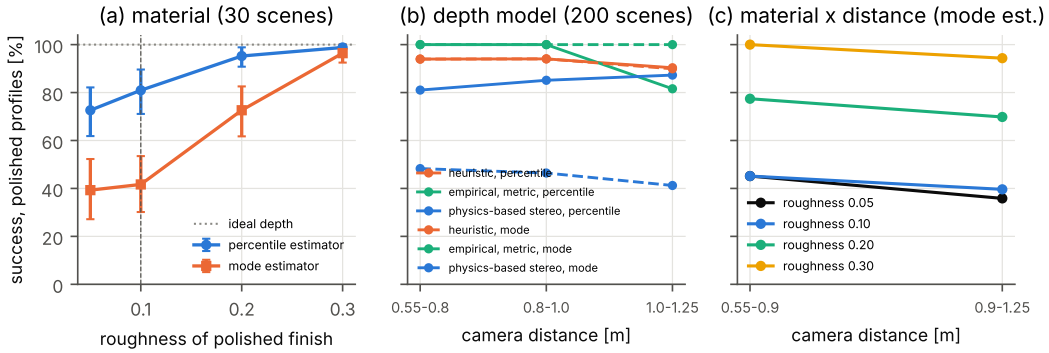


Fig. 7: (a) Success of the physics-based model on polished profiles vs. roughness (30 scenes, 84 profiles, 95% scene-bootstrap intervals). (b) Success on polished profiles vs. distance for the realistic models (solid: percentile, dashed: mode). (c) Roughness $\times$ distance, mode estimator.

the mean distance of the angle curves to ROBI from 9.6–13.2 to 13.4–21.7 points; the HDR sensor changed it by at most 0.1 points and on 200 scenes increased it on polished aluminium (14.1 vs. 12.3). All three are disabled.

4) **Anodized material.** Of three alternative anodized materials, roughness 0.30 reduced the distance to ROBI Zigzag from 9.4 to 8.2 points, but none reproduced the rise of holes at grazing incidence in ROBI, which XYZ-IBD does not show either (Fig. 4).
5) **Grasp estimator.** The first version took the median across the profile, which falls on one side of the central slot and caused a 4.1 mm lateral error even with ideal depth; we use the midpoint of the extent.

B. Interpretation and limitations

The answer to the task question is affirmative: replacing ideal or Gaussian depth by a D415 model validated on real data turns "every profile is grasped" into 15–55 points of failures on polished profiles and changes *which* profiles fail. This conclusion is robust to the estimator, to the variants of the physics-based model, to the grasp tolerance and to the statistical method; the *size* of the effect is not, since it varies by up to 41 points with the estimator, by up to 57 points with the polished roughness and, for the empirical model, with the transfer rule. Moreover, the outcome is not monotonic in realism: with the percentile estimator the physics-based model (85.2%) is less pessimistic than the empirical model with disparity transfer (65.4%). The two fail through different mechanisms, one through distance (Eq. (4)) and the other through orientation (Eq. (6)); realism is not a one-dimensional axis, and only a paired design can attribute an outcome to a model. We do not claim photometric realism of the rendered images; we claim depth-error statistics calibrated against real data. Learned noise generators [10] and learned stereo [11] are complementary: they can absorb residual artefacts, but only a physics-based chain validated on the depth error itself can predict artefacts for materials, finishes and orientations never recorded. Limitations: segmentation is assumed perfect; the grasp criterion is geometric, without gripper physics; materials were set by hand and checked against real depth statistics, not fitted to real infrared images [5]; the real reference sets contain other metal parts, ROBI at a shorter distance, and XYZ-IBD contributes 150 correlated frames from 15 scenes; no real grasp was performed.

VII. Riskiest Assumption and First Six Months

A. The riskiest assumption

The most damaging assumption is not that our sensor model is imperfect, but that the real distribution of depth errors on the parts and lighting of the target cell is contained in the simulated one. Three observations make it the largest risk. First, the same D415 varies strongly *between real datasets*: the invalid rate on metal is 1.9% in XYZ-IBD and 10.7–14.7% in ROBI (Tab. I), a factor of 5–8 for one sensor, comparable to the differences between our finishes and models; a model validated on ROBI may still fail on the parts of the target cell. Second, the predicted effect is not a property of the sensor model alone: it moves by up to 41 points with the downstream estimator, by up to 57 points with a roughness we set by hand (Fig. 7a), and depends on an undetermined distance-transfer rule (Sec. V-G). Every claim that "the simulation predicts the robot" inherits these unmeasured degrees of freedom. Third, in [10] a noise generator learned from only 80 short unpaired real recordings raised real-camera success from 0% to 95.7%: failures are dominated by unmodelled sensor artefacts, and the data-efficient remedy is to learn the residual from a small amount of real data.

B. Falsifiable predictions

The study yields three predictions that discriminate between the sensor models and can be tested on a real D415 without a robot:

P1 *Orientation* (physics-based model): on polished profiles the rate of height errors above 10 mm is higher at $\phi \approx 0°$ than at $90°$ (odds ratio 0.04–0.11). Test: one profile on a turntable, depth recorded every $15°$.

P2 *Distance* (empirical model, metric transfer): failures of the percentile estimator appear beyond $z^\star \approx 0.9\,\mathrm{m}$ (Eq. (4)). Test: the same profile recorded at 0.6–1.2 m and the measured $\hat{\sigma}_d$ inserted in Eq. (4).

P3 *Material*: the orientation effect of P1 vanishes as roughness increases (Fig. 7a). Test: P1 repeated on two finishes of measured roughness.

If P1 holds and P2 does not, the physics-based model is the right basis for training data; the converse would favour calibrated statistical models.

C. Plan for the first six months

Months 1–2: measure the real sensor on the real parts. Build a fixture with known poses, mount profiles in the finishes of the target cell (anodized, polished at measured roughness), record infrared pairs and raw depth with a D415 at 0.6–1.2 m and at controlled orientations to the baseline, extract the per-angle curves of holes and wrong pixels used here, and decide between P1–P3.

Months 2–4: fit what is currently set by hand. Estimate the material parameters and sensor thresholds by maximising the SRCC [8] on a held-out real set [5], fit the transfer exponent α at 0.6–1.2 m, and benchmark against learned stereo [11] and diffusion-based noise [10].

Months 4–6: transfer and report. Train perception and grasping on data from the validated sensor model, deploy on real grasps of the same parts and report the SRCC with confidence intervals [8], learning residual artefacts from a small set of unpaired real recordings [10]. Every change to the simulator is re-validated against real error statistics before it is used for training.

Use of AI Tools

An AI assistant was used for (i) literature retrieval and summarisation of related work on depth-sensor simulation and reflective-object datasets, (ii) drafting and debugging code for the stereo pipeline (census transform, semi-global matching), the Isaac Sim scene setup and the statistical analysis (cluster bootstrap, exact McNemar tests, GEE), and (iii) language editing. The experimental design, the interpretation of the results and all technical decisions are mine. I disagreed with the tool on two points. First, it suggested building the sensor simulation on a Blender-based pipeline as in [7]; I chose Isaac Sim, because I needed robot-oriented scene generation with built-in annotators for masks, normals and 6-D poses, and implemented the dot projector manually (Sec. VI-A). Second, it recommended enabling a measured infrared noise model and laser speckle for realism; I disabled both after they measurably increased the distance to the real ROBI data (Sec. VI-A), which I treated as the more reliable criterion.

VIII. Conclusion

A camera model that only adds Gaussian noise makes a grasp pipeline for aluminium profiles appear perfect. A physics-based active-stereo model of the D415, validated against real data with standard stereo metrics, predicts failures on polished parts and a specific, testable failure mechanism. The largest remaining uncertainty is not the sensor model but the scene it observes: the finish of the real parts and how far real errors carry across parts and distances.

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